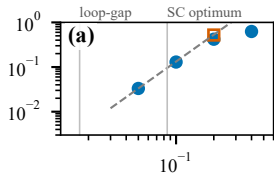
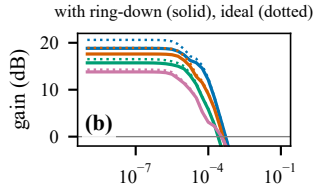


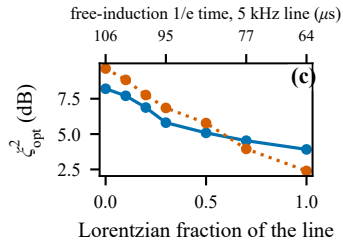
elimination loss (dB)



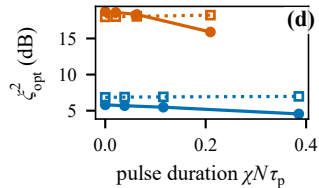
- exact, $N = 10$
- exact, $N = 20$
- $\propto (g\sqrt{N}/\Delta)^2$



- $\kappa/2\pi = 3$ kHz
- $\kappa/2\pi = 10$ kHz
- $\kappa/2\pi = 30$ kHz
- $\kappa/2\pi = 100$ kHz



- echo twist
- free twisting



- loop-gap, echo
- loop-gap, free
- SC operating point, echo
- SC operating point, free